No of Pages : 2 **Course Code : 23MX21**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, AUGUST / SEPTEMBER 2024

MCA Semester: 2

23MX21 SOFTWARE ENGINEERING

Time: 3 Hours **Maximum Marks:100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, (b) carries 7 marks each and (c) carries 15 marks each.
3. COCOMO is permitted to the students.

1. a) Write your comments on the following statement:

"With the addition of more people and program planners to Software development can help meet project deadlines (If lagging behind)."

b) i) For a project comprising many risks, would you recommend the prototyping or the spiral model? Justify the answer. (3)

ii) Tabulate the differences between Agile and Waterfall model (4)

c) For the following project description, generate complete estimates for total effort, project duration, and number of people needed using the COCOMO model.Progellis LLC., wants to produce a system that will perform computer-aided design for the automobile industry. They are a new company, and though they want to be the best in CAD systems, they are still, overall, a bit inexperienced.

Initial analysis of the problem leads to requirements calling for 3 major modules, with the following sizes: screen drawing 20 KLOC, object- base management 3500 LOC, algebra/numerical methods 1750 KLOC. Consider the following cost drivers: data base size – high, product complexity -very high, main storage -very high, execution time constraints –high, programmer capability - very high, programming language experience- very low. Use a=3.2, b=1.05, c=2.5 and d=0.38

2. a) Is it true that, the success of a project relies heavily on a crucial yet often overlooked phase is Requirement Gathering? Justify the answer.

b) i) Identify the functional and non-functional requirements for an e-commerce website. (4)

ii) While collecting the requirements for a department automation, one of the stakeholders mentioned the following requirement. Correct the problems in the following requirement, if any?

"During the final grade computation of a course. if any student scores a sufficiently low grade in a semester, then his/her parents would need to be informed . (3)

c) i) 1) Depict USE CASE Model for a Banking Software without ATM. (7)
2) Write a suitable USE CASE Story to illustrate the withdrawal of cash. (8)

(OR)

ii) 1) Design a suitable USE CASE Model to depict our college Library automation software [Online Public Access Catalogue]. (7)

2) Write a complete USE CASE Story to explain the student authentication to access our library. (8)

3. a) "A design solution is simple and understandable, if it is modular". How is modularity achieved?

b) Design a simple Class diagram for the ATM Banking software

c) i) Design a very neat activity model to explain the different steps in online shopping website using a standard mobile app.

(OR)

ii) Design a neat sequence and collaboration diagrams to explain the withdrawal of cash from ATM considering that, the customer is already authenticated.

4. a) "It is practically impossible to exhaustively test a program" Comment on this statement.

b) Apply Black Box method to test a module which is developed for customer authentication in an ATM Banking Software.

c) Given the following code, apply basis path testing to determine the number of test cases and generate the test case report in the standard template

```
1. Display "Enter any two numbers"
2. Read A, B
3. If (A>B)
Then
4. BIG = A
Else
5. BIG = B
6. Display "Largest Number is =", BIG
7. Display "DO you want to continue?"
8. Read Choice
9. If Choice="Yes"
Then
10. Goto statement 1
Else
11. EXIT
```

/END/

FD/RL